# PYTHON LIBRARY FILES IMPLEMENTATION

1 filtering data

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, 65, 90, 72]
})

print(df[df["Marks"] > 75])
```

```
  Name  Marks
0    A     85
2    C     90
```

2 filtering data with multiple conditions

```
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, 65, 90, 72],
    "Dept": ["ECE", "CSE", "ECE", "CSE"]
})

print(df[(df["Marks"] > 70) & (df["Dept"] == "ECE")])
```

```
  Name  Marks Dept
0    A     85  ECE
2    C     90  ECE
```

3 Sorting data

```python
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, 65, 90, 72]
})

print("Ascending:")
print(df.sort_values("Marks"))

print("\nDescending:")
print(df.sort_values("Marks", ascending=False))
```

```
Ascending:
  Name  Marks
1    B     65
3    D     72
0    A     85
2    C     90

Descending:
  Name  Marks
2    C     90
0    A     85
3    D     72
1    B     65
```

4 handling missing values

```python
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, np.nan, 90, np.nan]
})

print("Missing Values:")
print(df.isnull())

print("\nMissing Count:")
print(df.isnull().sum())
```

```
Missing Values:
    Name  Marks
0  False  False
1  False   True
2  False  False
3  False   True

Missing Count:
Name     0
Marks    2
dtype: int64
```

5 removing missing values

```python
import pandas as pd
import numpy as np

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, np.nan, 90, np.nan]
})

print("Before:")
print(df)

print("\nAfter Removing Missing Values:")
print(df.dropna())
```

```
Before:
  Name  Marks
0    A   85.0
1    B    NaN
2    C   90.0
3    D    NaN

After Removing Missing Values:
  Name  Marks
0    A   85.0
2    C   90.0
```

**6 removing duplicate data**

```python
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "B", "C"],
    "Marks": [85, 70, 70, 90]
})

print("Before:")
print(df)

print("\nAfter Removing Duplicates:")
print(df.drop_duplicates())
```

```
Before:
  Name  Marks
0    A     85
1    B     70
2    B     70
3    C     90

After Removing Duplicates:
  Name  Marks
0    A     85
1    B     70
3    C     90
```

## 7 Aggregation

```python
import pandas as pd

df = pd.DataFrame({
    "Dept": ["ECE", "ECE", "CSE", "CSE"],
    "Marks": [85, 90, 70, 80]
})

print(df.groupby("Dept")["Marks"].agg(["mean", "max", "min"]))
```

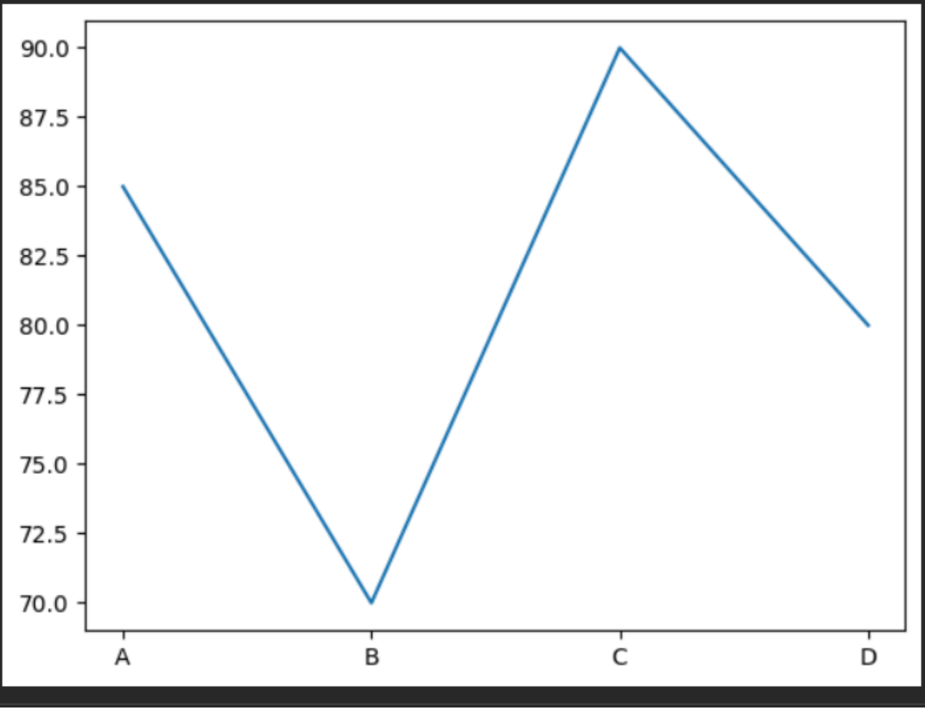
```
      mean  max  min
Dept
CSE   75.0   80   70
ECE   87.5   90   85
```

## 8 Data visualization

```python
import pandas as pd
import matplotlib.pyplot as plt

df = pd.DataFrame({
    "Name": ["A", "B", "C", "D"],
    "Marks": [85, 70, 90, 80]
})

plt.plot(df["Name"], df["Marks"])
plt.show()
```

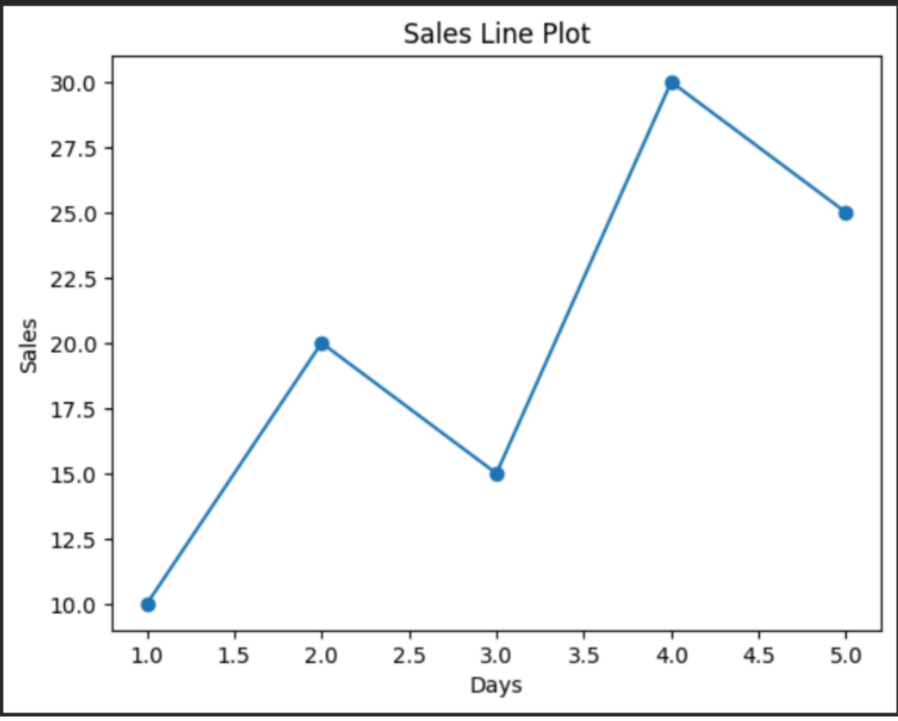


# 9 Line plot

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4, 5]
y = [10, 20, 15, 30, 25]

plt.plot(x, y, marker="o")
plt.xlabel("Days")
plt.ylabel("Sales")
plt.title("Sales Line Plot")
plt.show()
```

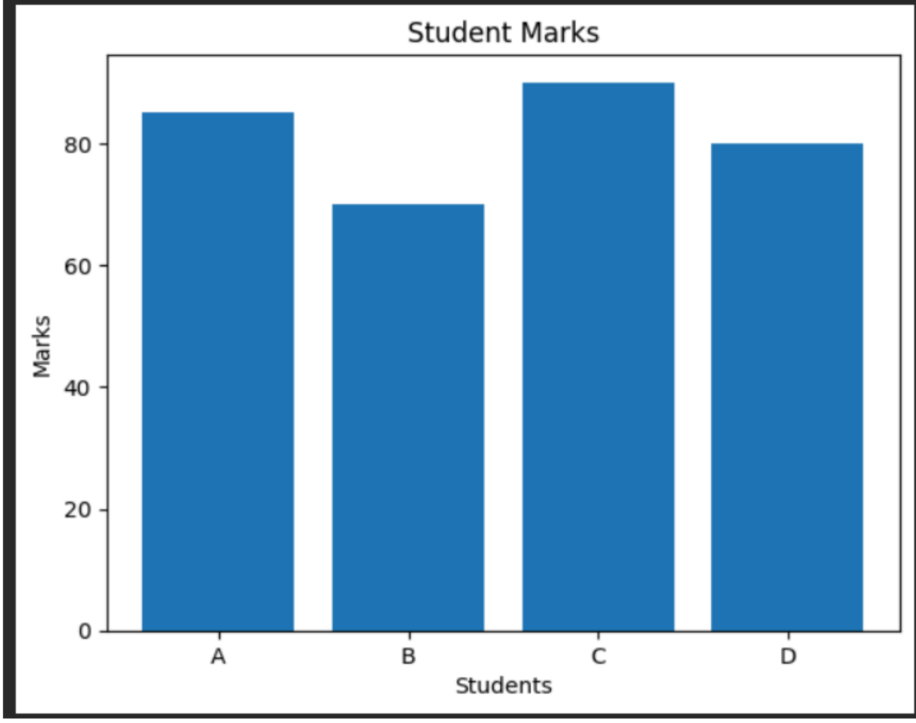


# 10 bar chart

```
import matplotlib.pyplot as plt

names = ["A", "B", "C", "D"]
marks = [85, 70, 90, 80]

plt.bar(names, marks)
plt.xlabel("Students")
plt.ylabel("Marks")
plt.title("Student Marks")
plt.show()
```

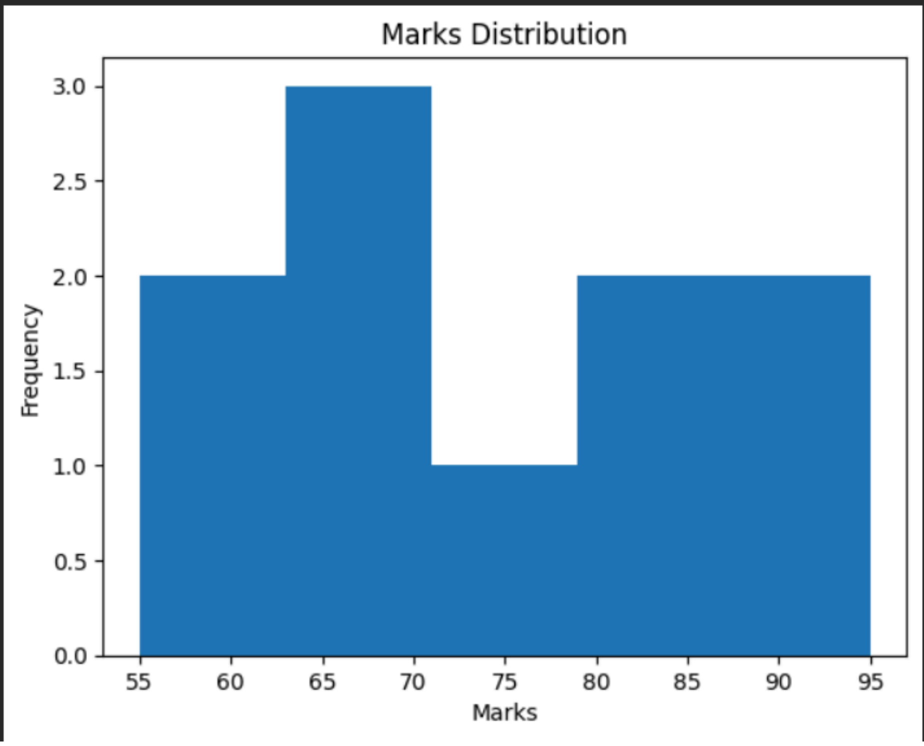


# 11 histogram

```
import matplotlib.pyplot as plt

marks = [55, 60, 65, 70, 70, 75, 80, 85, 90, 95]

plt.hist(marks, bins=5)
plt.xlabel("Marks")
plt.ylabel("Frequency")
plt.title("Marks Distribution")
plt.show()
```

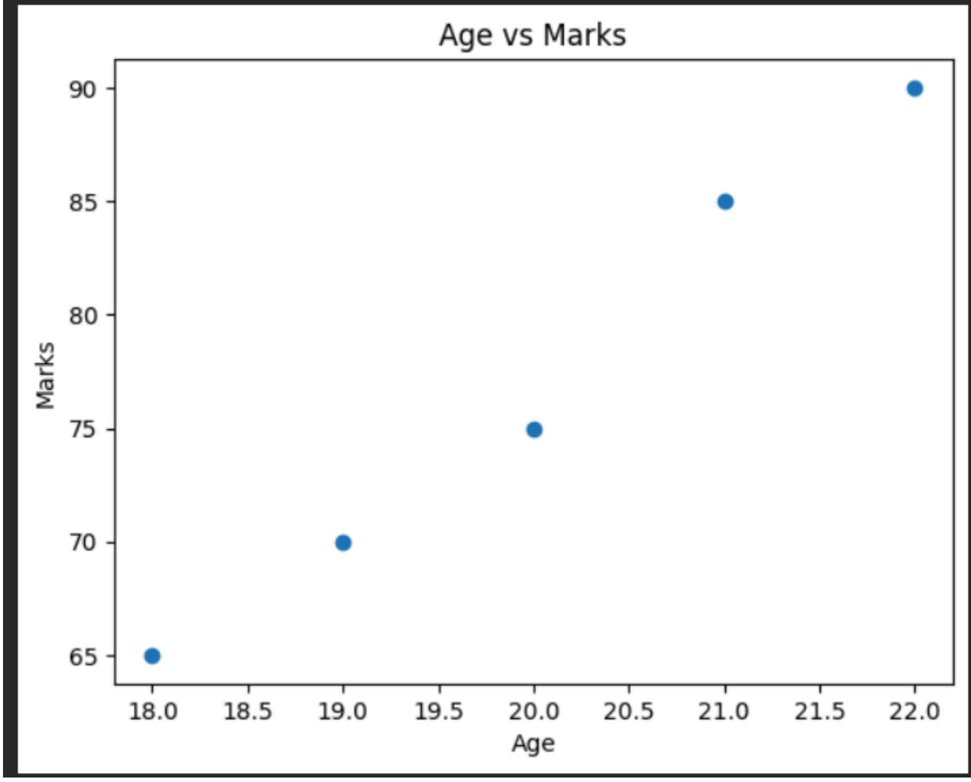


## 12 scatter plot

```python
import matplotlib.pyplot as plt

age = [18, 19, 20, 21, 22]
marks = [65, 70, 75, 85, 90]

plt.scatter(age, marks)
plt.xlabel("Age")
plt.ylabel("Marks")
plt.title("Age vs Marks")
plt.show()
```

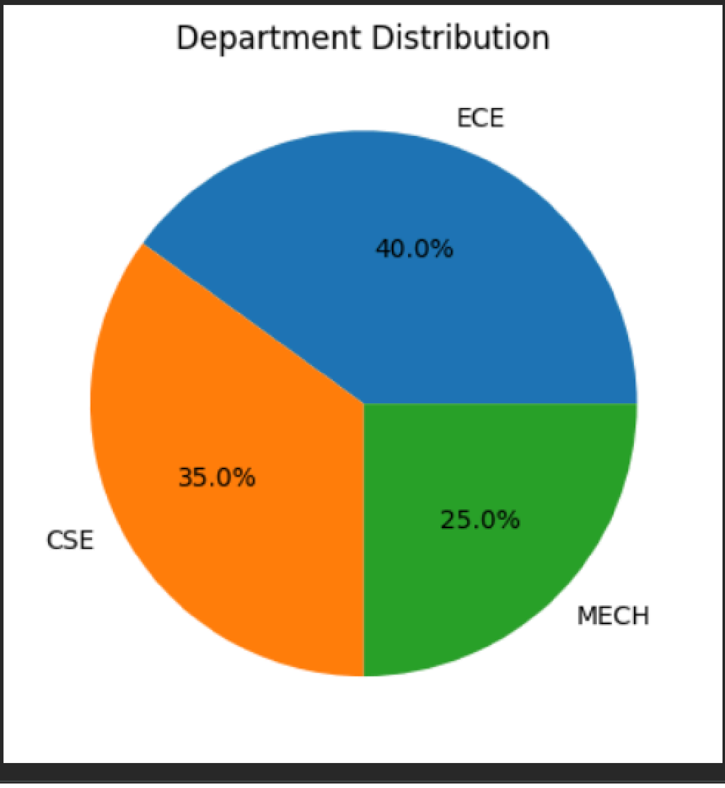


## 12 Piechart

```
import matplotlib.pyplot as plt

dept = ["ECE", "CSE", "MECH"]
students = [40, 35, 25]

plt.pie(students, labels=dept, autopct="%1.1f%%")
plt.title("Department Distribution")
plt.show()
```

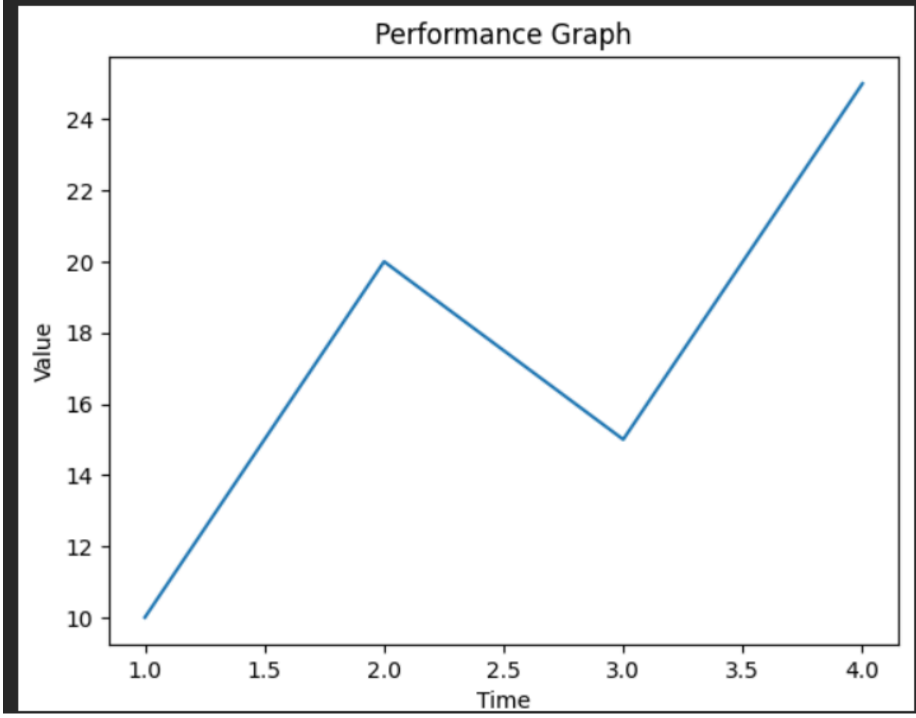


# 13 plot labels and titles

```
import matplotlib.pyplot as plt

x = [1, 2, 3, 4]
y = [10, 20, 15, 25]

plt.plot(x, y)
plt.xlabel("Time")
plt.ylabel("Value")
plt.title("Performance Graph")
plt.show()
```

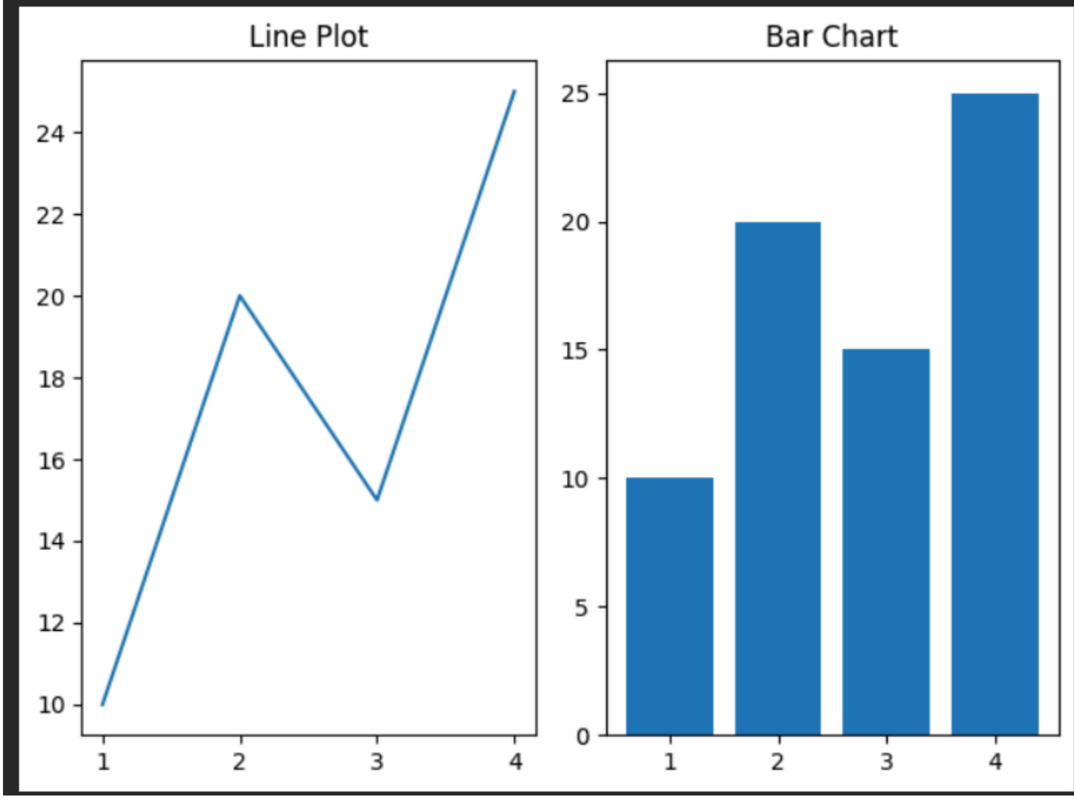


# 14 sub plot

```
plt.subplot(1, 2, 1)
plt.plot(x, y)
plt.title("Line Plot")

plt.subplot(1, 2, 2)
plt.bar(x, y)
plt.title("Bar Chart")

plt.tight_layout()
plt.show()
```



**15 reading and writing to a CSV file.**

```python
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B", "C"],
    "Marks": [85, 75, 90]
})

# Write CSV
df.to_csv("students.csv", index=False)

# Read CSV
data = pd.read_csv("students.csv")

print("CSV File:")
print(data)
```

```
CSV File:
  Name  Marks
0    A     85
1    B     75
2    C     90
```